Using the Geometry Viewer

START `python3 launch.py --remoll "/path/to/remoll"`

Run inside the package, with ROOT & Geant4 configured.
Windows: py -3 · Browser: 127.0.0.1:8784

1 Load files

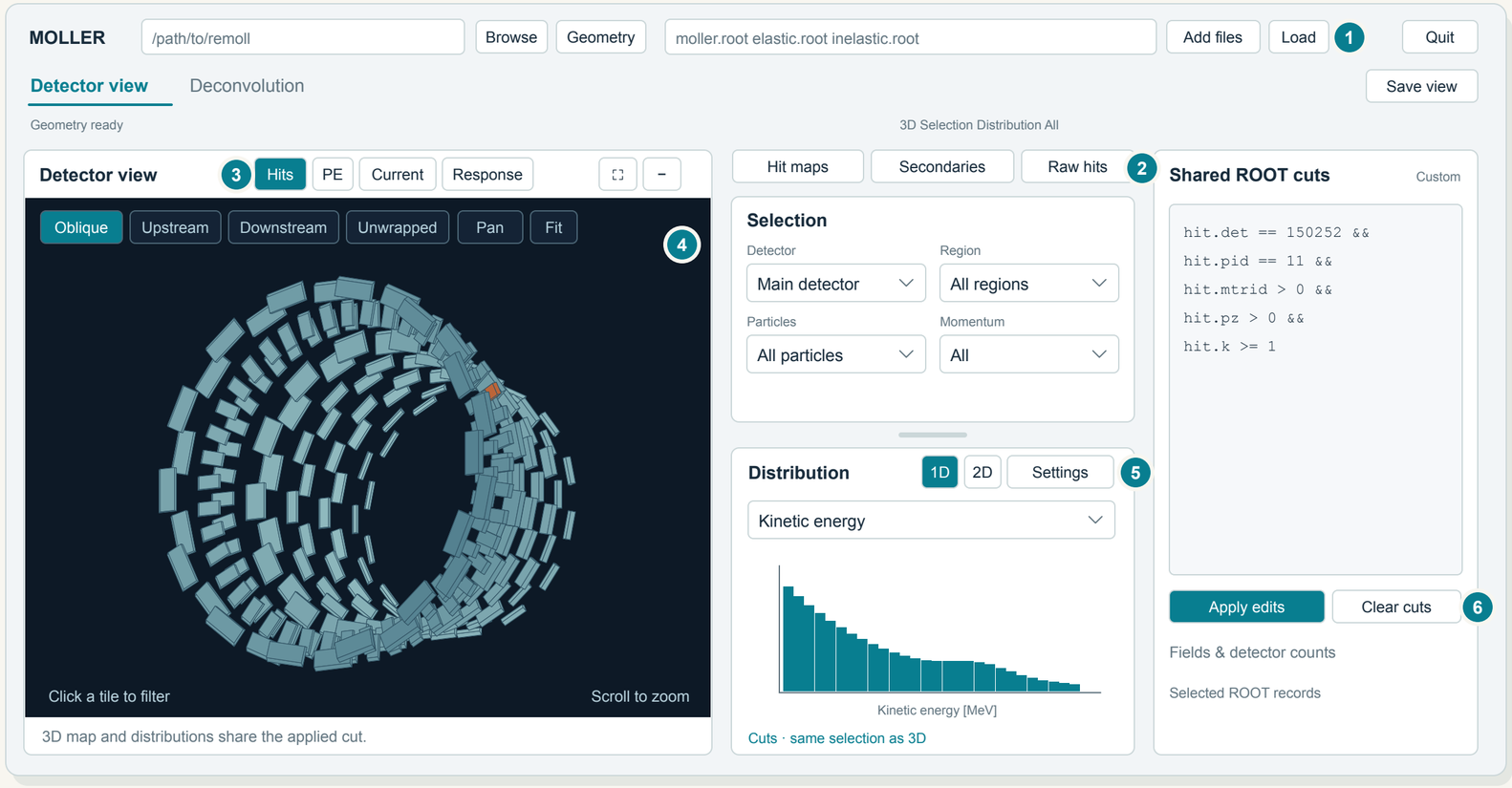
Browse → Geometry
Add files → Load
Or paste paths, one per line.

2 Choose a preset

Hit maps: main charged K ≥ 1 MeV.
Secondaries: adds mother ID > 0.
Raw hits: removes those defaults.
Then choose detector and region.

3 Choose the signal

Hits: selected record counts.
PE: expected photoelectrons/s.
Current: cathode current [nA].
Response: bundled or local files.



4 Move around

Drag to rotate; scroll to zoom.
Shift + drag to pan.
Double click to focus; Fit resets.
⛶ expands; − hides a panel.
Drag dividers to resize.

5 Shape the plot

Pick 1D or 2D, then a quantity.
Settings: labels, limits, log, grid.
ROOT histogram bins → Apply bins.
Save SVG exports the plot.

6 Edit the cut

Cuts → edit → Apply edits.
Changes affect map and plot.
Picking a preset replaces edits.

Controls illustrated · detector geometry from GDML · plotted data schematic

Try a cut

Replace the entire cut box → Apply edits. Examples use the current remoll detector IDs.

Ring 5 · Open

Charged records, kinetic energy ≥ 1 MeV

```
hit.det >= 150000 && hit.det < 152000 &&
hit.det%200 < 100 &&
(hit.det%100 == 51 || hit.det%100 == 52 ||
hit.det%100 == 53) &&
charged(hit.pid) && hit.k >= 1
```

One tile · Forward secondaries

Secondary electron records on tile 150252

```
hit.det == 150252 &&
hit.pid == 11 && hit.mtrid > 0 &&
hit.pz > 0 && hit.k >= 1
```

ShowerMax · Open

Forward records on the entrance planes

```
hit.det >= 73001 && hit.det <= 73028 &&
hit.det%4 == 3 && hit.pz > 0
```

`&&` both conditions `||` either condition `%` remainder `mtrid: 0 = primary; > 0 = secondary` k: MeV · x/y/z: mm

charged(hit.pid) excludes neutral particles. PE mode also enforces its model support; a custom cut does not override it.

7 Run deconvolution

Use Møller, ep elastic and ep inelastic runs,
with all 224 main tiles + 28 ShowerMax planes recorded.

Deconvolution → Check files → Run deconvolution

Set beam days, polarization and current in the lower corner.
Click a tile for correlations; use the tabs for matrices and uncertainties.
Display cuts do not change this fit. Uncertainties are forecasts.

Keep / stop

Save view: one local JSON download.
Quit or Ctrl+C stops the app.
Missing data? Read Check files / status.

Files stay on your computer.

Main PE: forward e±, K > 1.1 MeV · ShowerMax PE: forward supported particles.

Use All to restore hidden panels.